

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS8501 Theory of Computation

Type of Activity: Collaborative Learning

Topic: Context free grammar, Parse Trees and Ambiguity in Grammars and Languages

Date: 28-08-19

Objectives:

O1: To enable the students to learn and explore Context free grammar, Parse Trees and Ambiguity in Grammars and Languages.

Justification for chosen the topic:

Problems on the topic Context free grammar, Parse Trees and Ambiguity in Grammars and Languages were important and should be practiced. For better understanding of the following topics the collaborative tutorial was conducted to communicate the ideas for solving different problems.

Steps in Collaborative Learning:

- **Plan**-About group size and composition
- **Introduce**-Time limits and decision making within the group all need to be discussed before the learning.
- **Monitor**- Once students have begun to work in their groups, it is job of the teacher to monitor and observe. Teacher should be available to answer questions and provide clarification as needed.
- **Assess**- Judgment and solution explanations
- **Process**- Giving time to reflect their learning experience.

Activity Description:

Collaborative learning is an educational approach to teaching and learning that involves groups of students working together to solve a problem. Collaborative learning is based on the idea that learning is a naturally social act in which the participants talk among themselves.

The class was separated into eight groups (Team A to H). Each group has one leader. For each group set of problems were given. The team members and leader work together to solve a problem. Finally, the team leader explains the answers to other groups. The activity is conducted on CSE Seminar hall.



Reflection critique

Post Implementation:

- The activity encouraged the students to give more practices on construction of parse trees.
- It was a good experience for the students to learn different problems efficiently.

Observation Made:

Some of the students finding difficulties for solving Right Most Derivation from the given context free grammar and also how to find ambiguous grammar.

Steps taken for challenges:

- Additional problems were planned in ambiguous and rightmost derivation.
- Homework problem is given for the students to practice the concept of ambiguous and right most derivation better.



Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	2	2		2					2			

References:

<https://www.gdrc.org/kmgmt/c-learn/>

<https://www.edutopia.org/topic/collaborative-learning>

Faculty-Incharge

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HOD

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